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AutoVaultLK

User Manual

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

Table of Contents

1. Introduction....3
1.1 Purpose and Scope....3
1.2 System Overview....3
1.3 User Roles....3
1.4 Conventions....4
2. Getting Started and Account Management....4
2.1 Requirements and Access....4
2.2 Navigation....4
2.3 Buyer Registration and Sign-in....4
2.4 Password Recovery and Sessions....5
3. Buyer Functions....5
3.1 Searching for Vehicles....5
3.2 Vehicle Details....7
3.3 Saved Vehicles....7
3.4 Use on Mobile Devices....8
4. Dealer Functions....8
4.1 Registration and Verification....8
4.2 Dealer Portal and Dashboard....9
4.3 Managing Listings....10
4.4 Manual Listing....11
4.5 Bulk Upload....12
4.6 Business Details....14
5. Administrator Functions....14
5.1 Sign-in and Dashboard....14
5.2 User and Dealer Management....15
5.3 Upload Monitoring and Reports....15
5.4 Audit Logs and Dictionary Maintenance....16
6. Troubleshooting and Good Practice....16

1. Introduction

1.1 Purpose and Scope

This manual describes the operation of AutoVaultLK, a web-based marketplace for second-hand vehicles in Sri Lanka. It is intended for the three groups of people who use the system, namely buyers, dealers and administrators, and it explains each function from the viewpoint of the person carrying out a task. Programming details are outside the scope of this document.

1.2 System Overview

AutoVaultLK enables dealers to publish vehicle listings individually or by importing an entire inventory in one operation. It also enables buyers to locate vehicles through a search facility that accepts ordinary sentences, tolerates spelling errors and ranks the results by relevance. No technical details are required to use the system.

1.3 User Roles

Four categories of user are recognised, and the functions available to each are summarised in Table 1. A visitor is any person who uses the public pages without an account, and a buyer is a visitor who has created a free account. A dealer is a seller whose registration has been verified by an administrator. Dealers are either individuals or registered businesses, and only businesses may use the bulk upload facility. An administrator manages the platform; administrator accounts are created by the platform owner and cannot be registered through the public pages.

Table 1. Users of the system and the functions available to them.

Role	Account	Principal functions
Visitor	Not required	Browse the home page, search and filter vehicles, view vehicle details and contact dealers.
Buyer	Free, with email verification	All visitor functions, plus saving vehicles to a personal list.
Dealer	Verified by an administrator	Create and manage listings; business dealers may also import inventories in bulk.
Administrator	Created by the platform owner	Approve dealers, manage accounts, monitor uploads, run reports and maintain the dictionary.

1.4 Conventions

The names of buttons, tabs and fields are printed in bold exactly as they appear on screen. Screen captures are numbered consecutively, and all names, vehicles and figures that appear in them are sample data.

2. Getting Started and Account Management

2.1 Requirements and Access

The system is used through a current version of a standard web browser, such as Chrome, Edge, Firefox or Safari, on a computer, tablet or telephone, and requires an internet

connection. A valid email address is necessary for account confirmation and password recovery. Dealers additionally require either a business registration certificate in PDF, JPEG or PNG format of at most 5 MB, or a National Identity Card (NIC) number. The public pages open without signing in. Dealers sign in through the Dealer Portal, which is reached from the **Dealers** item of the main menu, whereas administrators use a separate sign-in page whose address is supplied by the platform owner and is not linked from the public pages.

2.2 Navigation

Every public page carries the same header, shown in Fig. 1. It contains the logo, the main menu (**Home**, **Browse**, **Just listed** and **Dealers**), a search icon that expands into a search box, a heart icon that opens the saved list, a **Sign in** link or, after sign-in, an account button, and an **Add listing** button that leads dealers to their portal. The home page itself presents a search panel with free-text, make and budget inputs, live counts of listed vehicles, verified dealers and makes, and a strip of the most recent listings that can be filtered by vehicle category. The dealer and administrator areas replace the header with a sidebar on the left, which also displays the signed-in email address and a **Sign out** button.

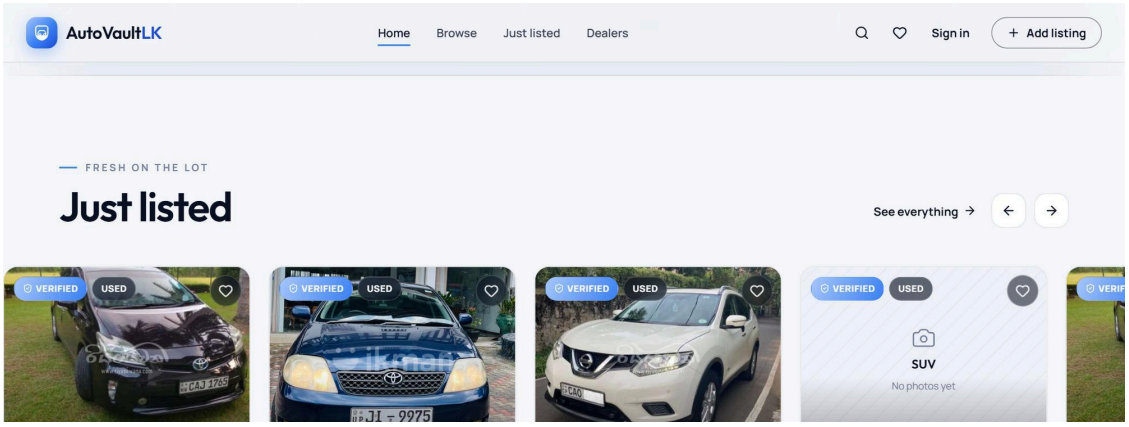


Fig. 1. Home page showing the header and the Just listed strip.

2.3 Buyer Registration and Sign-in

A buyer account is created from the **Create one** link on the sign-in page (Fig. 2). The form requests a full name, an email address and a password that is typed twice. A password must contain at least eight characters, including a lowercase letter, an uppercase letter and a digit, and a field that is incomplete or invalid is marked with a message beneath it. After submission, the system sends a verification email, and the account becomes usable only when the link in that email is opened. If the message does not arrive, a resend button is offered on the confirmation page and on the sign-in page. Once the address is verified, the buyer signs in with the email address and password and is returned to the page that was being visited, or to the search page.

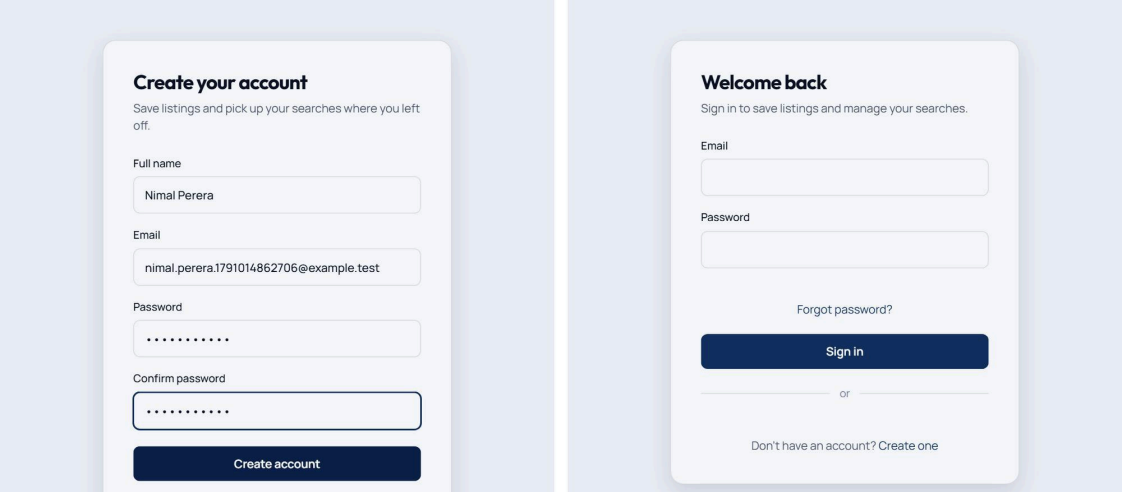


Fig. 2. Account creation form (left) and sign-in page (right).

2.4 Password Recovery and Sessions

A forgotten password is recovered through the **Forgot password?** link, which requests the email address and sends a reset link. The response is identical whether or not an account exists, which protects privacy. Only the most recent link remains valid, and each link expires after a limited period, which is about one hour by default. Sessions are renewed automatically while the system is in use, but a session that cannot be renewed ends with a request to sign in again. Repeated incorrect passwords cause sign-in to be suspended temporarily. Users of shared computers should select **Sign out** on completion.

3. Buyer Functions

3.1 Searching for Vehicles

Search is opened with **Browse**, the search icon or the home page search panel. Results are shown 20 per page together with the total count, and the layout comprises a filter panel on the left, the result cards and a sort menu at the top right (Fig. 3). Each card gives the photograph, the make and model, the price in Sri Lankan rupees (LKR) with a **Negotiable** label where applicable, the mileage, the fuel type, the transmission, the year of manufacture and the location. A **Verified** badge identifies listings from verified dealers, and a placeholder indicates that no photograph has been supplied.

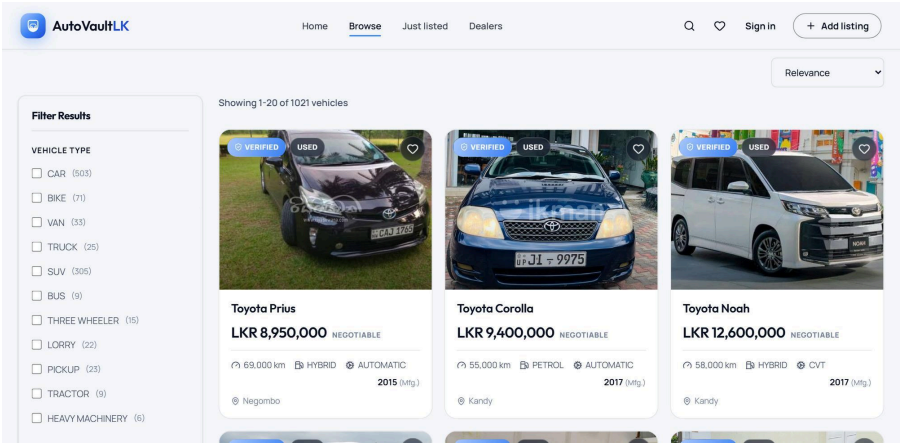


Fig. 3. Search page with the filter panel, sort menu and vehicle cards.

Natural-language search requires no filters. A sentence such as “Toyata Corrola under 8.5m deisel” is interpreted as a make, model, price limit and fuel type despite the spelling errors, and the interpretation appears as a chip above the results (Fig. 4). Words that cannot be interpreted are reported in a notice and ignored, and the results can then be refined with the filters.

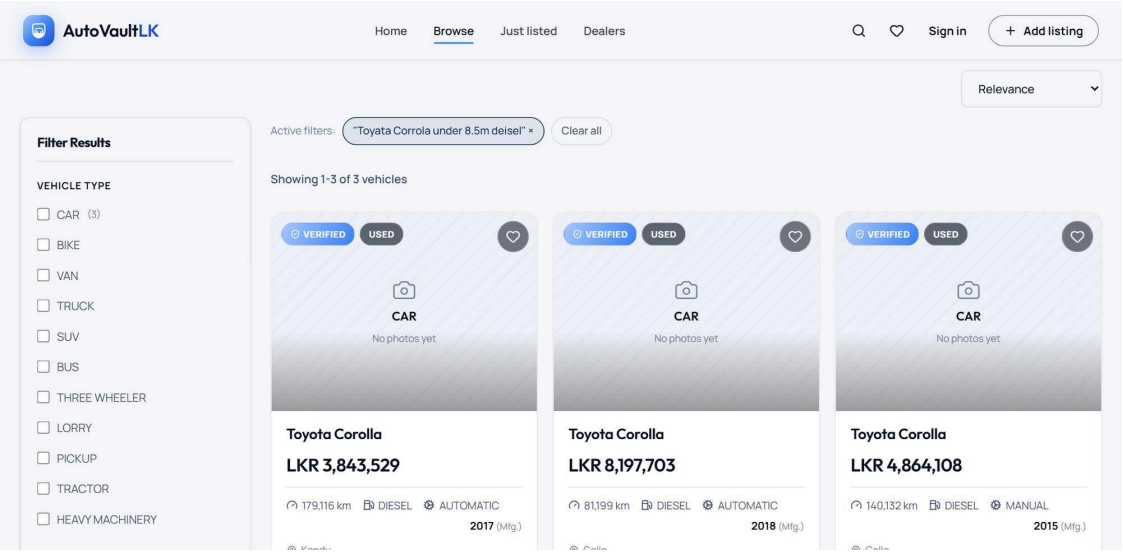


Fig. 4. Natural-language search with the interpreted query shown as a chip.

The filter panel narrows the results by vehicle type, make, model, price, year, condition, maximum mileage, fuel type, transmission, district, body type and number of previous owners, together with four yes-or-no options: negotiable price only, verified dealers only, confirmed registration year only and sunroof. Filters that apply only to certain vehicles appear when they are relevant, namely seats and drive type for passenger vehicles, engine class for motorcycles and load capacity for trucks and lorries. The count beside each option shows how many vehicles match it. Every applied filter is shown as a removable chip, and **Clear all** restores the full list. Results can be ordered by relevance, most recent, price in either direction, newest year or lowest mileage. When a search finds nothing, the system relaxes it step by step, for example by removing a mileage range, and reports what was relaxed. If nothing matches even after relaxation, an explanatory message and a **Clear all filters** button are displayed.

3.2 Vehicle Details

Selecting a card opens the detail page (Fig. 5). The left side presents the photograph gallery, the dealer's description and the specification list, which includes the type, year, mileage, condition, fuel, transmission, colour, engine capacity, previous owners and any type-specific attributes such as seats or load capacity. The summary panel on the right states the price, key facts, location and dealer, with a green **Verified** label for verified dealers and a **Call** button when the dealer has provided a telephone number. A **Similar vehicles** strip at the foot of the page suggests comparable listings, and a vehicle that has been sold or removed produces a “Listing not available” page. Viewing, negotiation and payment are arranged directly with the dealer, and the vehicle and its documents should be inspected before any payment is made.

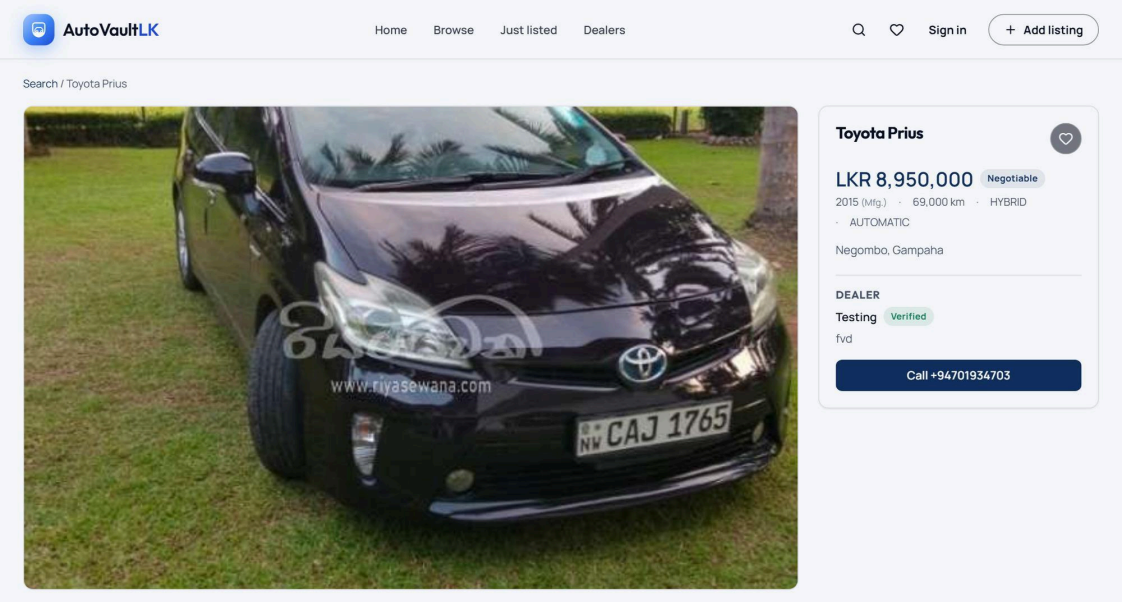


Fig. 5. Vehicle detail page.

3.3 Saved Vehicles

A signed-in buyer saves a vehicle by selecting the heart icon on a card or on the detail page, and removes it by selecting the icon again. A message at the top right confirms each action, and a visitor who selects the icon is directed to the sign-in page. The saved list, which is opened from the heart icon in the header, displays the saved vehicles as cards (Fig. 6), and a vehicle removed from the list disappears immediately.

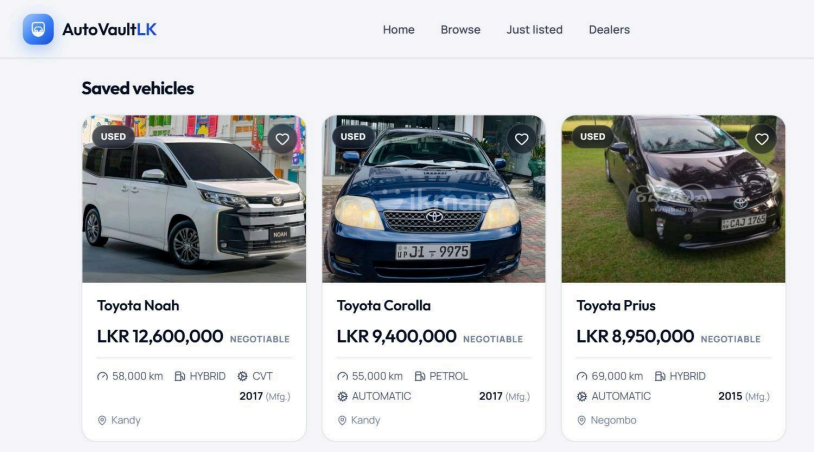


Fig. 6. Saved vehicles page.

3.4 Use on Mobile Devices

The interface adapts to small screens, as illustrated in Fig. 7. The main menu collapses behind a menu button, the results are shown in a single column, and the filter panel slides over the page when the **Filters** button is selected and is closed with **Close filters**. All buyer functions described above are available on a telephone.

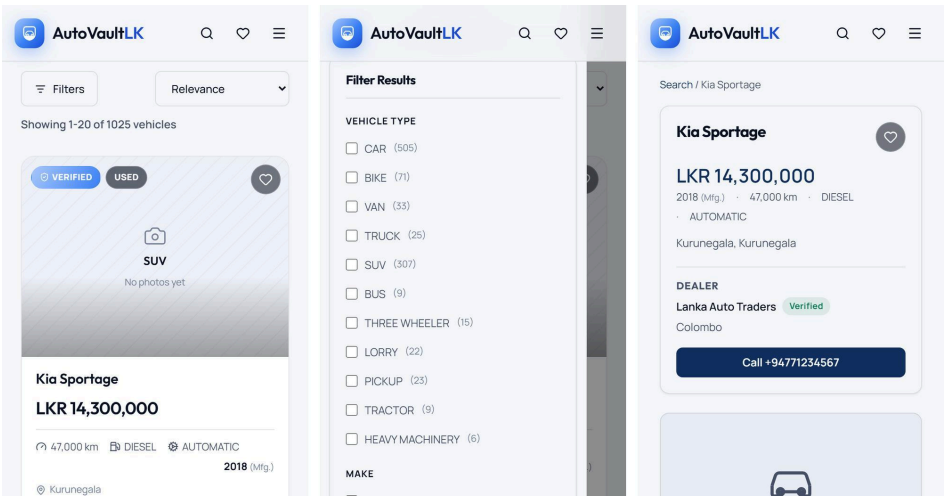


Fig. 7. Mobile layout: search results (left), filter panel (centre) and vehicle detail (right).

4. Dealer Functions

4.1 Registration and Verification

Dealer registration is a four-step wizard that is started from **Register as dealer** on the home page or from the Dealer Portal sign-in page (Fig. 8). The first step collects the company name, the dealer type, the business registration number, the address and the city. Business dealers must also upload a registration certificate, whereas individual dealers enter an NIC number consisting of nine digits followed by V or X, or of twelve digits, and may omit the registration number. The second step records the contact name and the telephone number, which is entered without the leading zero beside a country code. The third step sets the sign-in email address and password, and the fourth presents a summary for confirmation before the **Create Dealer Account** button is used. Each step is validated before the wizard proceeds.

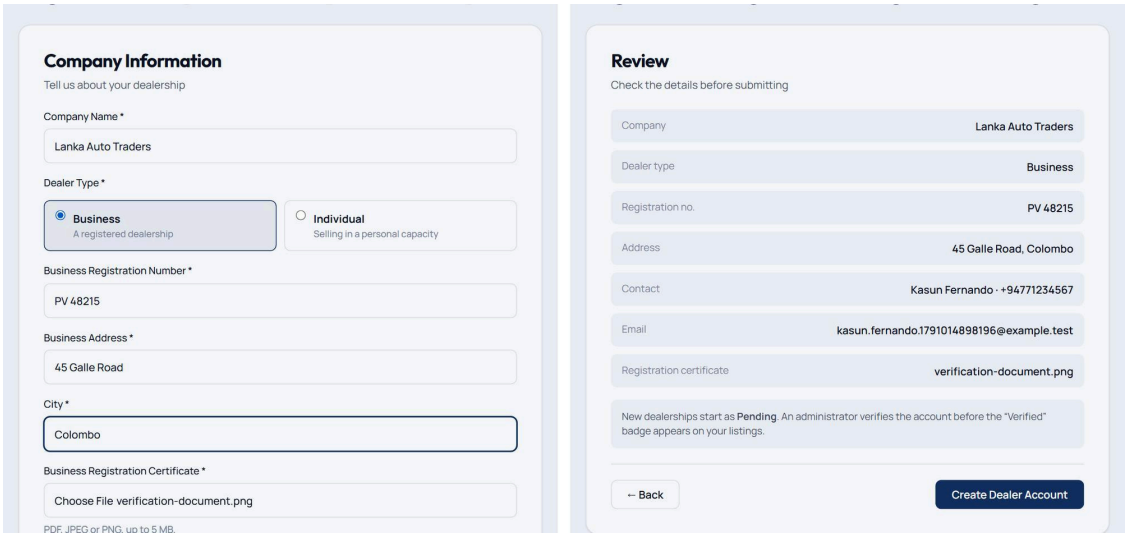


Fig. 8. Dealer registration: company information (left) and review (right).

A verification email is then sent and, once its link has been opened, the dealer may sign in at once. Until an administrator has reviewed the application, however, the portal shows only a

Verification pending notice and no menu. After approval, the full portal opens and a **Verified** badge appears on the dealer's listings. If the application is rejected, the reason entered by the administrator is displayed together with a form for correcting the details, and **Resubmit for review** returns the application to the pending state.

4.2 Dealer Portal and Dashboard

The sidebar of the portal offers **Dashboard**, **My listings**, **Manual listing** and **Business details** and, for business dealers only, **Bulk upload** and **Upload history**. The dashboard (Fig. 9) summarises the stock through tiles for live, pending review, draft, rejected and archived listings and for the total asking price of live listings. It also shows the five most recent listings and a chart of the inventory by make.

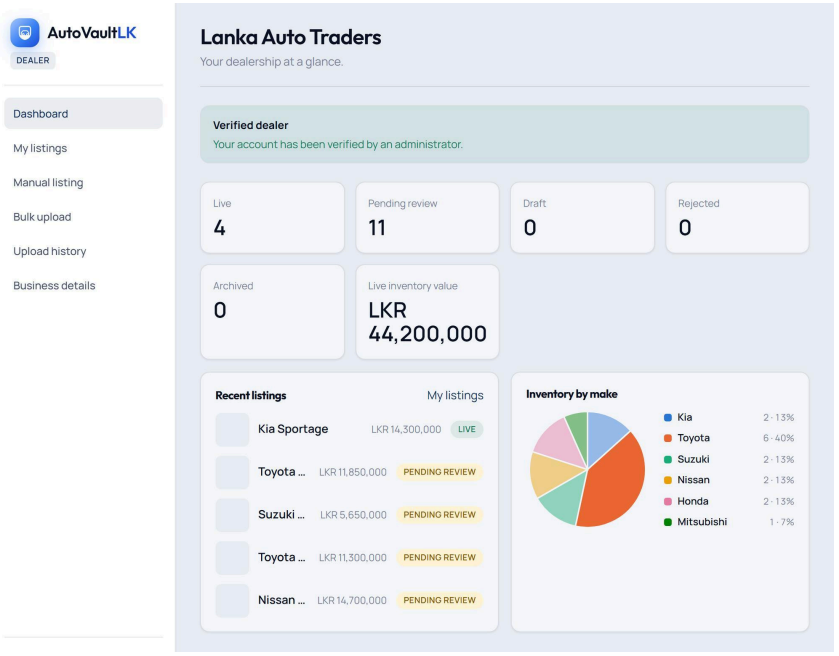


Fig. 9. Dealer dashboard.

4.3 Managing Listings

The **My listings** page lists every vehicle owned by the dealer with its type, condition, year, price, mileage and status. A listing passes through the statuses shown in Fig. 10, where each arrow is an action available to the dealer. Vehicles brought in by bulk upload begin as Pending review and are invisible to buyers until approved, whereas manual listings created by a verified dealer are published immediately as Live. A live listing can be archived, which removes it from search, and an archived listing can be restored. A listing that has never been published may be deleted permanently, but a published listing can only be archived, because buyers may already have saved it.

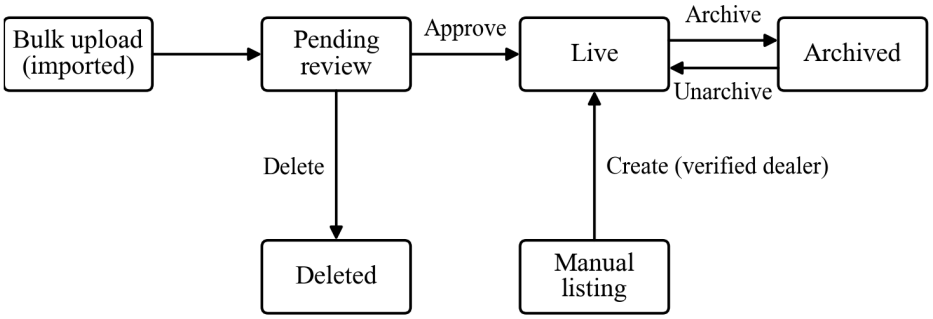


Fig. 10. Life cycle of a listing.

Listings awaiting review are highlighted by a banner that states their number (Fig. 11). Beneath each row, **Show field details** explains how every imported value was interpreted and labels it as parsed, matched or corrected automatically, so that the corrections can be checked before publication, while **Show specs & photos** displays the stored specifications and the photographs matched to the vehicle. A single listing is published with **Approve**, and several listings are published together by ticking their check boxes and selecting **Approve selected**. **Edit** opens the listing form. Searchable changes made there are re-indexed automatically, and photographs can be replaced or removed individually. **Archive** and **Unarchive** change the availability after a confirmation prompt, and **Delete** is offered only where it is permitted.

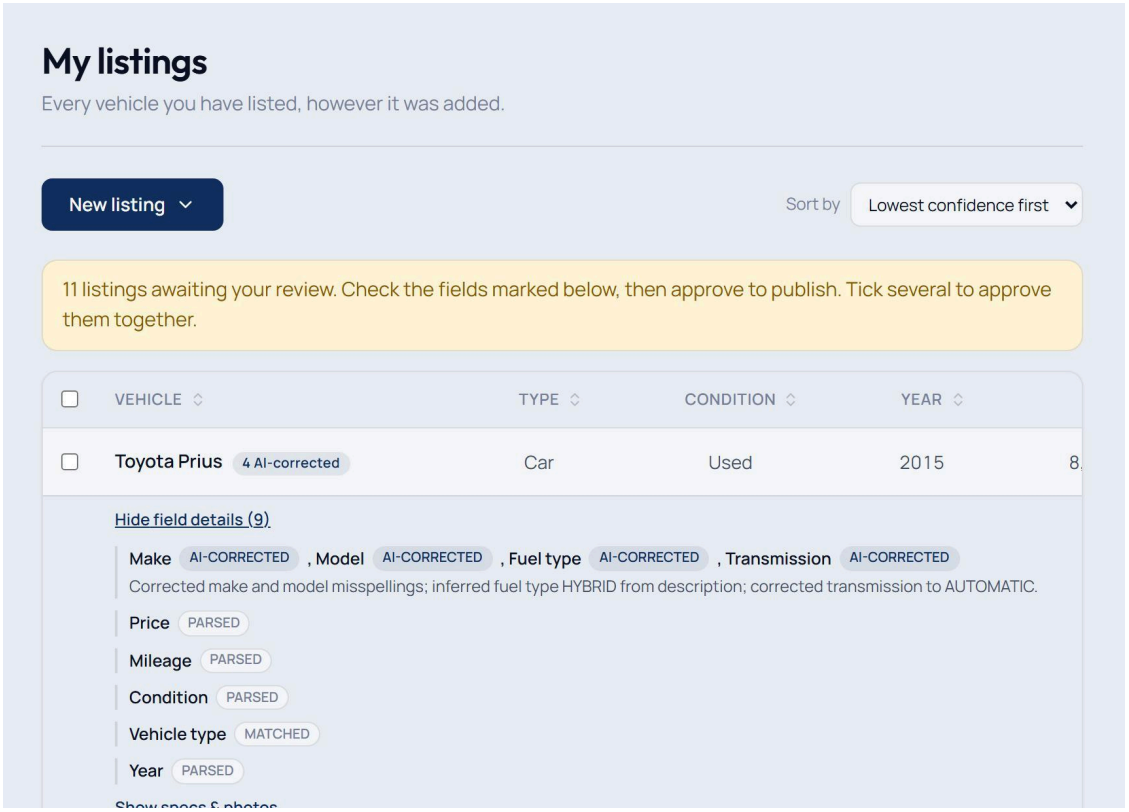


Fig. 11. My listings with the field details of an imported vehicle.

4.4 Manual Listing

A single vehicle is added through **Manual listing** (Fig. 12). The vehicle type is chosen first, because it determines which further sections are displayed. The mandatory fields are the type, make, model, year of manufacture, price in LKR, mileage, fuel type, transmission, colour, engine capacity, number of previous owners and district, while the registration year, condition, city, body type, registration and chassis numbers and a negotiable-price option are optional. Further sections collect the attributes of cars, motorcycles, vans and buses, and trucks, lorries and pickups, and an equipment section, which is hidden for motorcycles and three-wheelers, records items such as a sunroof or a reverse camera. A free-text description and up to ten photographs complete the form. The photographs must be in JPEG, PNG or WebP format and must not exceed 8 MB each, and the first photograph becomes the main image. Missing or invalid fields are highlighted when **Create listing** is selected. A slow connection may delay the confirmation. For that reason, **My listings** should be checked before a listing is submitted again, so that duplicates are avoided.

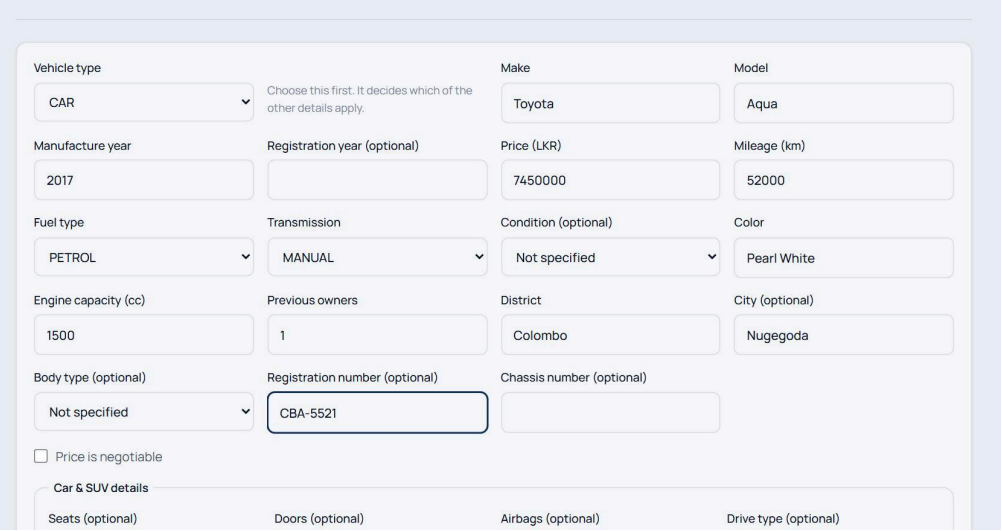


Fig. 12. Manual listing form.

4.5 Bulk Upload

Bulk upload, which is available to business dealers, imports an entire inventory in one operation and follows the six steps shown in Fig. 13, namely preparation of the files, upload, automatic processing, the result report, dealer review and final approval. The inventory is supplied as a UTF-8 CSV file or as a JSON array of flat objects, up to 25 MB in size, with one row per vehicle. Thirteen columns are mandatory, namely make, model, year, price, mileage, fuel type, transmission, colour, engine capacity, number of owners, district, condition and vehicle type, and all other columns, such as the registration number, body type and equipment, are optional. Values are interpreted flexibly, so that “Rs. 3,500,000”, “3.5M” and “45,000 km” are accepted, and misspelled makes and models are corrected when they are close to a known entry. A template file, a table of all columns and a list of recognised makes and models are available from the buttons at the top of the upload page (Fig. 14).

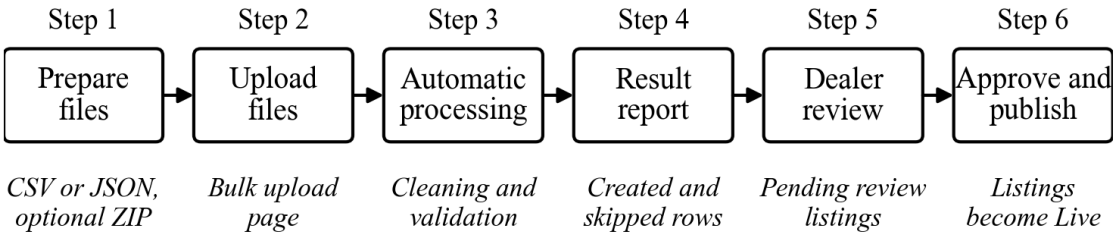


Fig. 13. Bulk upload workflow.

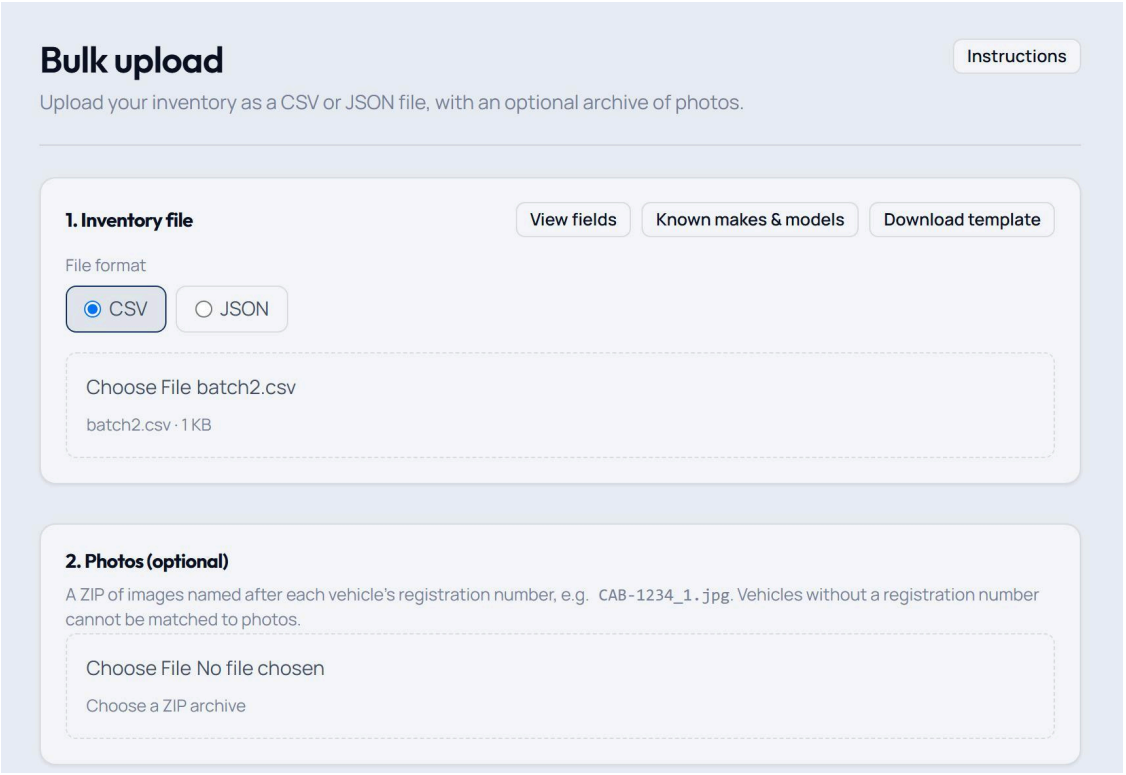


Fig. 14. Bulk upload page with an inventory file selected.

An optional ZIP archive of photographs of up to 250 MB may accompany the inventory. Each image must be named after the registration number of its vehicle, with a numeric suffix for further images, for example CAB-1234.jpg and CAB-1234_2.jpg. Images whose registration number is absent from the inventory are skipped, and vehicles without a registration number cannot be matched and must be given photographs through the edit form. After **Upload inventory** is selected, a status page reports the progress and, finally, the numbers of rows loaded and skipped (Fig. 15). A job that ends as **Completed with skipped rows** is a successful import in which the valid rows have been created, and each skipped row is listed with its row number, the reason and the data received. Typical reasons are a value outside the permitted range, a missing mandatory value, an unrecognised make and a registration number that is already listed. The skipped rows can be downloaded as a CSV file, corrected, and uploaded again after the **Row** and **Reason** columns have been removed. The created listings are pending review and must be approved as described in Section 4.3, and every upload is retained under **Upload history**.

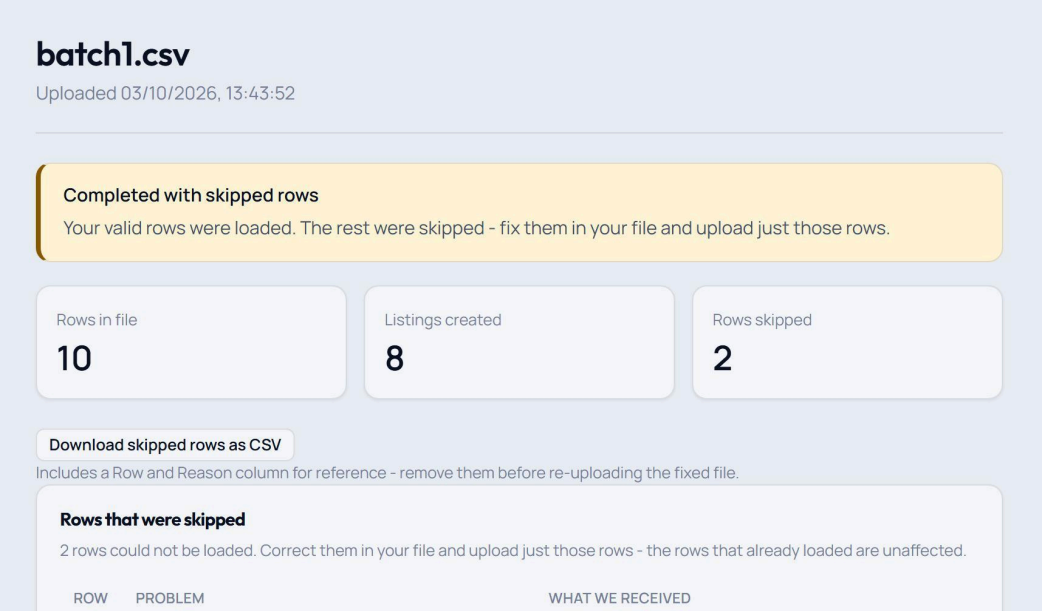


Fig. 15. Result of a bulk upload with skipped rows.

4.6 Business Details

The **Business details** page allows the contact name, company name, business registration number (business dealers only), address, city and contact number to be edited. The **Save changes** button becomes active once a field has been changed. The email address and the dealer type cannot be altered after registration.

5. Administrator Functions

5.1 Sign-in and Dashboard

Administrators sign in on a dedicated page, and any other user who attempts to open the console is returned to the sign-in page. The sidebar provides **Dashboard**, **Users**, **Uploads**, **Reports**, **Audit logs** and **New vehicle types**. The dashboard (Fig. 16) reports the numbers of live listings, users, dealers and pending approvals, together with three 30-day trend charts for new listings, new users and submitted uploads, breakdowns of listings and uploads by status, notification delivery and the most recent administrative actions.

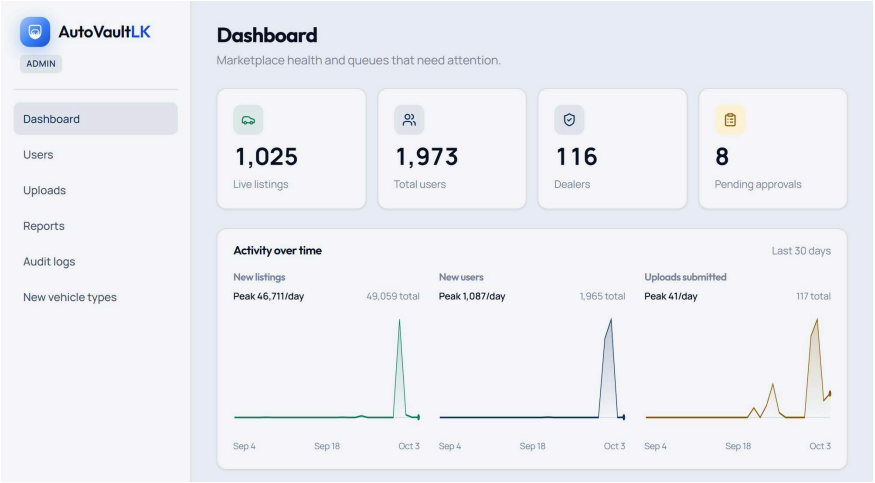


Fig. 16. Administrator dashboard.

5.2 User and Dealer Management

The **Users** page lists all accounts with their role, status, dealer details and join date, and its **Pending dealers** tab restricts the list to applications awaiting review (Fig. 17). The **View document** link opens the certificate uploaded by a business dealer for inspection. **Approve** verifies the dealer, whereas **Reject** requests a reason that is displayed to the dealer, who may then correct and resubmit the application. **Deactivate** prevents an account from signing in once the prompt has been confirmed, and administrators cannot deactivate themselves or one another. Each of these actions is recorded in the audit log.

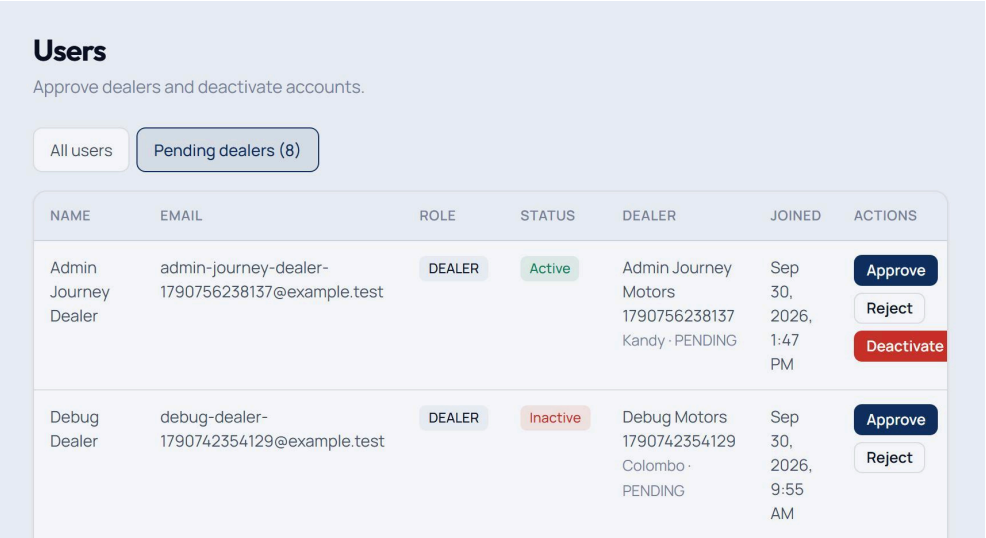


Fig. 17. Pending dealers tab of the Users page.

5.3 Upload Monitoring and Reports

The **Uploads** page lists the bulk upload jobs of all dealers with their file name, status, valid and total record counts and creation time, and it can be filtered by status. For a job with invalid rows, **View reasons** shows the same skipped-row report as the dealer sees, which allows administrators to assist dealers who encounter problems. The **Reports** page produces a summary for a chosen date range, comprising the numbers of accounts created, upload jobs and records processed, the job error and partial rates, and charts of listing status, row acceptance and job outcome. The report can be saved as a PDF document with **Download PDF**.

5.4 Audit Logs and Dictionary Maintenance

The **Audit logs** page presents the latest 200 matching administrative events with the time, action, entity, acting administrator and network address, and it can be searched by action, entity type, actor and date range (Fig. 18, upper part). The **New vehicle types** page collects the makes that dealers typed but the dictionary did not recognise, with the number of occurrences, the closest known make and sample rows (Fig. 18, lower part). **Add as alias** treats the text as a misspelling of the closest make, so that later uploads resolve to it automatically. **Add as new make** adds a genuinely new make with an editable display name, and **Dismiss** removes noise such as typing errors. Entries that recur across several dealers are the most likely to be genuine.

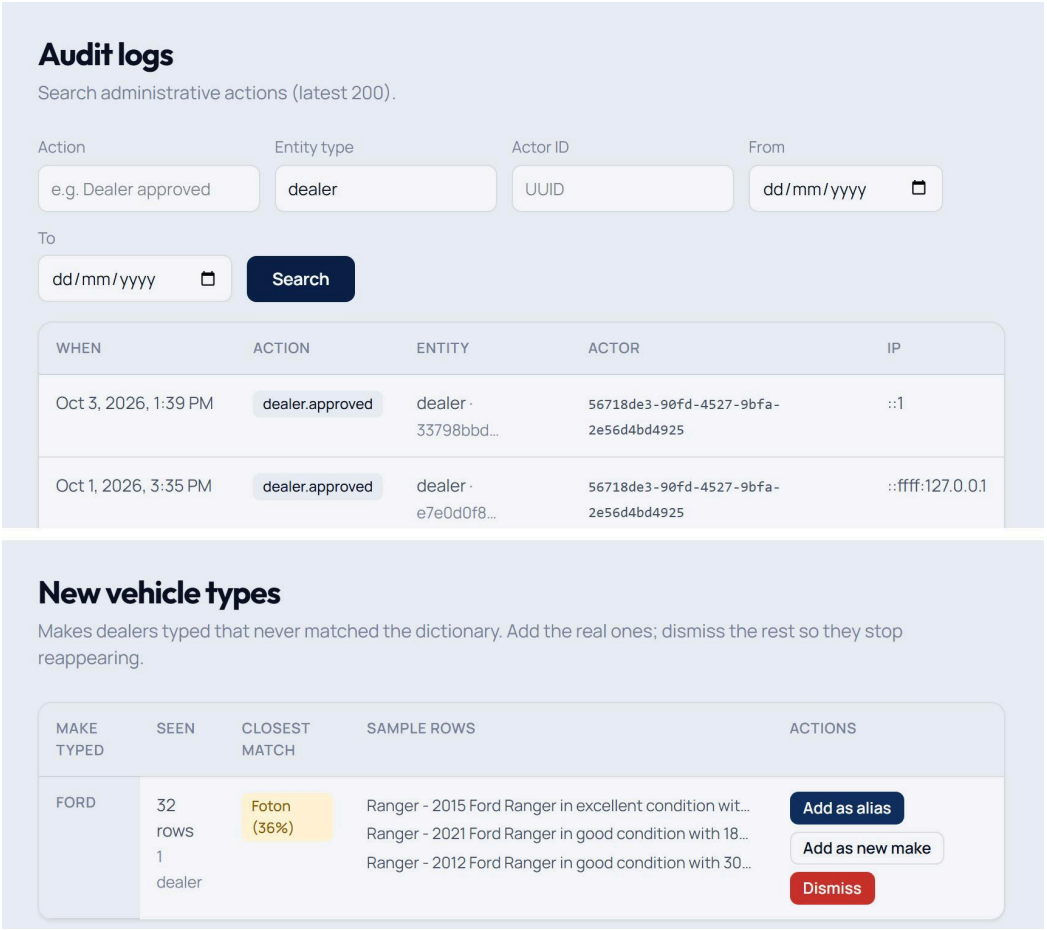


Fig. 18. Audit logs page (upper) and New vehicle types page (lower).

6. Troubleshooting and Good Practice

Most difficulties concern accounts. A missing verification email is normally found in the spam folder or is requested again through the resend button, and a verification or reset link that fails has usually expired or has been replaced by a newer one. A dealer who can sign in but sees no menu is either pending or rejected. In the first case the decision should be awaited, and in the second the application should be corrected and resubmitted. Dealer registration cannot proceed past the first step until a certificate has been uploaded, and the contact number must be entered without the leading zero.

Search problems are usually resolved by removing a filter chip or widening the price range, and a listing that has disappeared has been sold or archived. A bulk upload that fails entirely is generally caused by missing mandatory columns, by unrecognised makes in every row or by registration numbers that are already listed, and the skipped-row report states which applies. Photographs that do not attach should be checked for file names that match the registration number exactly.

Because the platform does not handle payments, buyers should inspect a vehicle and its registration documents in person before paying. Passwords should be unique and never shared, emails that were not requested should be ignored, and dealers should keep their contact numbers current so that buyers can reach them.